

Eigen Values & Eigen Vectors

Characteristic Polynomial = $|A - \lambda I|$

A is $n \times n$ matrix. I is $n \times n$
 λ is scalar no.

Characteristic Equation is $|A - \lambda I| = 0$

eg.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 \\ 3 & 4-\lambda \end{bmatrix}$$

Ch. eqⁿ = $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 \\ 3 & 4-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(4-\lambda) - 6 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda - 2 = 0$$

Note: Addn. of e-values = trace of A
 = addn of diagonal elements of A

multiplication of e-values = $|A|$ //

$$A = \begin{bmatrix} 1 & 3 \\ 0 & 4 \end{bmatrix}$$

Ch. eqⁿ $|A - \lambda I| = 0$

$$\lambda^2 - (5)\lambda + |A| = 0$$

$$\lambda^2 - 5\lambda + 4 = 0$$

$$\lambda = 4, 1$$

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -2 & 1 & 2 \\ 0 & 0 & -5 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Cayley-Hamilton Thm.

$$f(\lambda) = 0$$

$$f(A) = 0$$

verify C.H. Thm & hence find A^{-1}

$$A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$\text{Ch Eqn } |A - \lambda I| = 0$$

$$\lambda^3 - (1+3+1)\lambda^2 + [(3+3+1) - (-2+0+0)]\lambda - |A| = 0$$

$$\lambda^3 - 5\lambda^2 + 9\lambda - 1 = 0$$

To verify C.H. Thm.

$$\text{i.e. To verify } A^3 - 5A^2 + 9A - I = 0$$

$$\text{LHS} = A^3 - 5A^2 + 9A - I$$

$$A^3 - 5A^2 + 9A - I = 0$$

multiplying by A^{-1}

$$A^2 - 5A + 9I - A^{-1} = 0$$

$$A^{-1} = A^2 - 5A + 9I$$

$$= \begin{bmatrix} -13 & 42 & -2 \\ -11 & 9 & 10 \\ 10 & -22 & -3 \end{bmatrix} - 5 \begin{bmatrix} -1 & 12 & -4 \\ -4 & 7 & 2 \\ 2 & -8 & 1 \end{bmatrix} + 9 \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -13 & 42 & -2 \\ -11 & 9 & 10 \\ 10 & -22 & -3 \end{bmatrix} - \begin{bmatrix} -5 & 60 & -20 \\ -20 & 35 & 10 \\ 10 & -40 & 5 \end{bmatrix} + \begin{bmatrix} 9 & 18 & -18 \\ -9 & 27 & 0 \\ 0 & -18 & 9 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad A^3 - 5A^2 + 9A - I = 0$$

Hence Cayley-Hamilton Thm verified.

as we've By C.H. Thm.

$$A^3 - 5A^2 + 9A - I = 0$$

$$2A^{-1} = A^2 - 3A + 2I$$

$$A^{-1}(A^3 - 5A^2 + 9A - I) = 0$$

$$A^2 - 3A + 2I = 0$$

$$A^2 - 5A + 9I - A^{-1} = 0$$

$$A^{-1} = A^2 - 5A + 9I$$

$$= \begin{bmatrix} -1 & 12 & -4 \\ -4 & 7 & 2 \\ 2 & -8 & 1 \end{bmatrix} - \begin{bmatrix} 5 & 10 & -10 \\ -5 & 15 & 0 \\ 0 & -10 & 5 \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

$$\text{By C.H. Thm } A^3 - 5A^2 + 9A - I = 0$$

$$A(A^3 - 5A^2 + 9A - I) = 0$$

$$A^4 - 5A^3 + 9A^2 - A = 0$$

$$A^4 = 5A^3 - 9A^2 + A$$

$$A^4 = 5 \begin{bmatrix} -13 & 42 & -2 \\ -11 & 9 & 10 \\ 10 & -22 & -3 \end{bmatrix} - 9 \begin{bmatrix} -1 & 12 & -4 \\ -4 & 7 & 2 \\ 2 & -8 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -55 & 104 & 24 \\ -20 & -15 & 32 \\ 32 & -40 & -23 \end{bmatrix}$$

find.

$$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$$

where $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ | Also find A^{-1} using C.H.Tm

u. eqn $|A - \lambda I| = 0$

$$\lambda^3 - (5)\lambda^2 + [(2+2+4) - (0+1+0)]\lambda - |A| = 0$$

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

By Cayley-Hamilton thm

$$A^3 - 5A^2 + 7A - 3I = 0$$

multiplying by A^{-1} .
 $A^2 - 5A + 7I - 3A^{-1} = 0$

$$3A^{-1} = A^2 - 5A + 7I$$

$$= \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} - \begin{bmatrix} 10 & 5 & 5 \\ 0 & 5 & 0 \\ 5 & 5 & 10 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$$

Consider

$$\begin{array}{r} \lambda^5 + \lambda \\ \lambda^3 - 5\lambda^2 + 7\lambda - 3 \overline{) \lambda^8 - 5\lambda^7 + 7\lambda^6 - 3\lambda^5 + \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda + 1} \\ - \lambda^8 + 5\lambda^7 - 7\lambda^6 + 3\lambda^5 \\ \hline \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda + 1 \\ - \lambda^4 + 5\lambda^3 - 7\lambda^2 + 3\lambda \\ \hline \lambda^2 + \lambda + 1 \end{array}$$

Division algorithm.

$$\lambda^8 - 5\lambda^7 + 7\lambda^6 - 3\lambda^5 + \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda + 1 = (\lambda^3 - 5\lambda^2 + 7\lambda - 3)(\lambda^5 + \lambda) + \lambda^2 + \lambda + 1$$

$$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I = (A^3 - 5A^2 + 7A - 3I)(A^5 + A) + A^2 + A + I$$

$$= 0 + A^2 + A + I \quad \text{By C.H. Tm from ①.}$$

$$= \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix} //$$

$$\therefore = 0$$

4 C.H. $A^3 - 4A^2 - 20A - 35I = 0$ — ①

$$\begin{array}{r} \lambda^4 + \lambda \\ \lambda^3 - 4\lambda^2 - 20\lambda - 35 \overline{) \lambda^7 - 4\lambda^6 - 20\lambda^5 - 34\lambda^4 - 4\lambda^3 - 20\lambda^2 - 33\lambda + 1} \\ - \lambda^7 + 4\lambda^6 + 20\lambda^5 + 35\lambda^4 \\ \hline \lambda^4 - 4\lambda^3 - 20\lambda^2 - 33\lambda + 1 \\ - \lambda^4 + 4\lambda^3 + 20\lambda^2 + 35\lambda \\ \hline 2\lambda + 1 \end{array}$$

$$(\lambda^7 - 4\lambda^6 - 20\lambda^5 - 34\lambda^4 - 4\lambda^3 - 20\lambda^2 - 33\lambda + 1) = (\lambda^4 + \lambda)(\lambda^3 - 4\lambda^2 - 20\lambda - 35) + 2\lambda + 1$$

$$(A^7 - 4A^6 - 20A^5 - 34A^4 - 4A^3 - 20A^2 - 33A + I) = (A^4 + A)(A^3 - 4A^2 - 20A - 35I) + (2A + I)$$

$$= 0 + 2A + I$$

$$= 2 \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 6 & 14 \\ 8 & 5 & 6 \\ 2 & 4 & 3 \end{bmatrix}$$

Find

$$A^7 - 4A^6 - 20A^5 - 34A^4 - 4A^3 - 20A^2 - 33A + I$$

$$A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$$

Eigen Values :

The roots of ch. eqⁿ $|A - \lambda I| = 0$ are called as Eigen values / Ch. values / Ch. roots / latent values / proper values of matrix A .

Note ✓

2x2 $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Ch. eqⁿ $|A - \lambda I| = 0$

$$\lambda^2 - (a+d)\lambda + |A| = 0$$

3x3 $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

Ch. eqⁿ $|A - \lambda I| = 0$

$$\lambda^3 - (a+e+i)\lambda^2 + [(ae+ei+ia) - (bd+g+fh)]\lambda - |A| = 0$$

$$\left[\begin{array}{cc|ccc} 1 & 2 & & & \\ 2 & -2 & & & \end{array} \right] \quad \left[\begin{array}{ccc} -2 & -9 & 5 \\ -5 & -10 & 7 \\ -9 & -21 & 14 \end{array} \right]$$

$$\lambda + \lambda - 6 = 0$$

$$\lambda = -3, 2$$

$$\lambda^3 - 2\lambda^2 - \lambda + 2 = 0$$

$$\lambda = -1, 2, 1$$

$$|A - \lambda I| = 0$$

Eigen Values

Note:-

addition of e. values = trace of A

Characteristic Values

multiplier of e. values = $|A|$

Roots.

Latent Roots
Proper Values.

$$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 5 & -1 \\ 0 & 0 & -2 \end{array} \right]$$

Ch. eqⁿ $|A - \lambda I| = 0$

$$\lambda^3 - (4)\lambda^2 + [(5-10-2)-(0)]\lambda - |A| = 0$$

$$\Rightarrow \lambda^3 - 4\lambda^2 - 7\lambda + 10 = 0$$

$$\lambda = 5, 1, -2$$

Note:- For a lar matrix / Diagonal matrix - eigen values are diagonal elements of matrix.

Eigen Vectors :-

A be $n \times n$ matrix & λ be an eigen value of A then there exist a non-zero vector x such that

$$Ax = \lambda x$$

i.e. $Ax - \lambda x = 0$

$$(A - \lambda I)x = 0$$

Here x is an eigen vector corr-to eigen value λ

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix}$$

$$\text{Ch. eqn } |A - \lambda I| = 0$$

$$\lambda^2 - (3)\lambda + |A| = 0$$

$$\lambda^2 - 3\lambda - 4 = 0 \Rightarrow \lambda = 4, -1$$

Eigen values are 4 & -1.

To find Eigen vector $x \neq 0$ such that.

$$(A - \lambda I)x = 0$$

$$\begin{bmatrix} 1-\lambda & 3 \\ 2 & 2-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

for $\lambda = 4$

$$\begin{bmatrix} -3 & 3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x - 2y = 0 \Rightarrow x = y$$

$$\text{put } y = 1 \Rightarrow x = 1 \quad x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{for } \lambda = -1 \quad \text{from (1)} \quad \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{aligned} 2x + 3y &= 0 \\ x &= -\frac{3y}{2} \end{aligned}$$

$$\text{for } y = 2 \Rightarrow x = -3 \quad x_2 = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

Note: If x is eigen vector corr. to λ then kx is also an e. vector corr. to λ

For distinct e. values we'll get distinct (l.i) e. vectors.

$A = \begin{bmatrix} 8 & -2 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & -1 \end{bmatrix}$
 as $\lambda = 1, 2, 3$ are distinct $\Rightarrow$ e. values $\Rightarrow AM = IM = I$ for all e. values $\Rightarrow A$ is diagonalizable.

Ch. eqⁿ: $|A - \lambda I| = 0$

Ch. eqⁿ: $\lambda^3 - (8-3+1)\lambda^2 + [(-24-3+8) - (-32-6+8)]\lambda - |A| = 0$

$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\lambda = 1, 2, 3$ are eigenvalues of A

To find eigenvector $X \neq 0$ s.t. $(A - \lambda I)X = 0$

$\begin{bmatrix} 8-\lambda & -2 & -2 \\ 4 & -3-\lambda & -2 \\ 3 & -4 & 1-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ — ①

put $\lambda = 1$ in ①

$\begin{bmatrix} 7 & -2 & -2 \\ 4 & -4 & -2 \\ 3 & -4 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 $4x - 4y - 2z = 0$
 $3x - 4y + 0 = 0$
 $\frac{x}{-4-2} = \frac{-y}{4-2} = \frac{z}{3-4} = t$
 $\Rightarrow \frac{x}{-8} = \frac{-y}{6} = \frac{z}{-4} = t \Rightarrow x = -8t, y = -6t, z = -4t$
 $t = -1 \Rightarrow x_1 = \begin{bmatrix} 8 \\ 6 \\ 4 \end{bmatrix}$

$a_1x + a_2y + a_3z = 0$
 $b_1x + b_2y + b_3z = 0$
 $\frac{x}{a_1/a_3} = \frac{-y}{a_2/a_3} = \frac{z}{a_3/a_3} = t$
 $\frac{a_2/a_3}{b_2/b_3} = \frac{a_1/a_3}{b_1/b_3}$
 By Cramer's Rule

put $\lambda = 2$ in ①

$\begin{bmatrix} 6 & -2 & -2 \\ 4 & -5 & -2 \\ 3 & -4 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 $4x - 5y - 2z = 0$
 $3x - 4y - z = 0$
 $\Rightarrow \frac{x}{-3} = \frac{-y}{2} = \frac{z}{-1} = t$
 $x = -3t \Rightarrow y = -2t, z = -t$
 $t = -1 \Rightarrow x_2 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

$(1^{st} \& 3^{rd} \text{ row is same})$
 By Cramer's Rule

put $\lambda = 3$ in ①

$\begin{bmatrix} 5 & -2 & -2 \\ 4 & -6 & -2 \\ 3 & -4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 $4x - 6y - 2z = 0$
 $3x - 4y - 2z = 0$
 $\Rightarrow \frac{x}{-6-2} = \frac{-y}{4-2} = \frac{z}{3-4} = t$
 $\Rightarrow \frac{x}{-8} = \frac{-y}{6} = \frac{z}{-4} = t$
 $t = 1/2 \Rightarrow x_3 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

$\Rightarrow \frac{x}{-8} = \frac{-y}{6} = \frac{z}{-4} = t$
 $t = 1/2 \Rightarrow x_3 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$
 eigen values are $5, -3, -3$
 for $\lambda = 5$ $x_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

Char eqⁿ $|A - \lambda I| = 0$

$\lambda^3 - (-1)\lambda^2 + [(-2+0+0) - (4+3+12)]\lambda - |A| = 0$

$\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$

$\lambda = 5, -3, -3$

To find $x \neq 0$ s.t. $(A - \lambda I)x = 0$

$\begin{bmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ — (1)

$\lambda = 5$

$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

By Cramer's Rule

$2x - 4y - 6z = 0$

$-x - 2y - 5z = 0$

$x = 8t, y = 16t, z = -8t$

$t = 1/8, x_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

put $\lambda = -3$ in (1)

$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$x + 2y - 3z = 0$

$x = -2y + 3z$

$y = 0$

$z = 1$

$x = 3$

$x_2 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$

$y = 1$

$z = 0$

$x = -2$

$x_3 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$

$\begin{bmatrix} -2y + 3z \\ y \\ z \end{bmatrix}$

$\begin{bmatrix} -2y \\ y \\ 0 \end{bmatrix} + \begin{bmatrix} 3z \\ 0 \\ z \end{bmatrix}$

$y \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$

$$A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{bmatrix} \quad \text{Ch- eqn } |A - \lambda I| = 0$$

$$\lambda^3 - 5\lambda^2 + [(12 - 6 - 8) - (6 - 6 - 10)]\lambda - |A| = 0$$

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

$\lambda = 1, 2, 2$
eigen values

To find $x \neq 0$ such that $(A - \lambda I)x = 0$

$$\begin{bmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -5 & -2-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

$$\lambda = 1 \quad \left\{ \begin{bmatrix} 3 & 6 & 6 \\ 1 & 2 & 2 \\ -1 & -5 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\} \quad \lambda = 2$$

$$x + 2y + 2z = 0$$

$$-x - 5y - 3z = 0$$

By Cramer's Rule

$$\frac{x}{4} = \frac{+y}{+1} = \frac{z}{-3} = t$$

$$x = 4t, y = t, z = -3t \quad \text{for } t=1 \quad x_1 = \begin{bmatrix} 4 \\ 1 \\ -3 \end{bmatrix}$$

$$x + y + 2z = 0$$

$$-x - 5y - 4z = 0$$

By C.R.

$$\frac{x}{6} = \frac{-y}{+2} = \frac{z}{-4} = t$$

$$x = 6t, y = 2t, z = -4t$$

$$t = 1/2 \quad x_2 = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$$

$$(3, 2) = 3(1, 0) + 2(0, 1)$$

3+3

If we have ^{one} repeated eigen value (repeated twice)

$$[A - \lambda I]x = 0$$

Solve

If we get only one eqⁿ after solve $[A - \lambda I]x = 0$



Two eigen vectors

If we get two eqⁿ. after solving $[A - \lambda I]x = 0$

↓ Cramer's Rule

One e. vector

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\lambda = 2, 2, 2$$

$$(A - \lambda I)x = 0$$

$$\begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & 2-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix}$$

$$x_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad x_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad x_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Algebraic Multiplicity (AM)

If eigen value λ of matrix A is repeated k times then AM of λ is k

Geometric Multiplicity (GM)

If 's' is no. of linearly independent eigen vectors corr. to eigen value λ then GM of $\lambda = s$

$$AM \geq GM$$

Similar Matrices:

If A & B square matrices then B is said to be similar to A if there exist non-singular M such that $B = M^{-1}AM$

Diagonalisable Matrix:-

Square matrix A is said to be diagonalisable if there exist non-singular matrix M such that $D = M^{-1}AM$ (i.e. A is similar to D)

Thm: If λ is eigen value of A

& $AM = GM$ for all eigen values of A then A is diagonalisable

Note:

① If all eigen values are distinct then $AM = GM = 1$
 $\therefore$ matrix is diagonalisable

② A is diagonalisable $\Rightarrow D = M^{-1}AM$
Here D is diagonal matrix
 M is transforming / Modal matrix

To find M & D .
To find D

eg. If $\lambda_1, \lambda_2, \lambda_3$ are e-values
$$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

To find M corr. e-vectors as columns of M .

eg. If x_1, x_2, x_3 are e.v. corresponding to e-values $\lambda_1, \lambda_2, \lambda_3$

$$M = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}$$

① $\int$

$$A = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$$

Ch. eqⁿ $|A - \lambda I| = 0$

$$\lambda^3 - 2\lambda^2 + [(4 - 12 - 3) - (-12)]\lambda - 0 = 0$$

$$\lambda^3 - 2\lambda^2 + \lambda = 0$$

$$\lambda(\lambda^2 - 2\lambda + 1) = 0$$

$$\lambda = 0, 1, 1$$

To find e-vector $x \neq 0$ such that

$$(A - \lambda I)x = 0$$

$$\begin{bmatrix} 1-\lambda & -6 & -4 \\ 0 & 4-\lambda & 2 \\ 0 & -6 & -3-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda = 1 \quad \begin{bmatrix} 0 & -6 & -4 \\ 0 & 3 & 2 \\ 0 & -6 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$3y + 2z = 0 \Rightarrow y = -\frac{2}{3}z$$

Sumⁿ.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{2}{3}z \\ z \end{bmatrix}$$

$$= x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ -\frac{2}{3} \\ 1 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ -\frac{2}{3} \\ 1 \end{bmatrix}$$

for $\lambda = 1$ $AM = 2$ $GM = 2$ $\Rightarrow AM = GM$

$$\lambda = 0 \quad \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x - 6y - 4z = 0$$

$$4y + 2z = 0$$

$$-12 + 12 = 0$$

By Cramer

$$\frac{x}{4} = \frac{-y}{2} = \frac{z}{4} = t$$

$$\Rightarrow x = 4t, y = -2t, z = 4t$$

$$t=1 \quad x_3 = \begin{bmatrix} 4 \\ -2 \\ 4 \end{bmatrix} \quad AM = GM = 1$$

as $AM = GM$ for all eigenvalues $\Rightarrow A$ is diagonalizable

i.e. there exist invertible matrix M such that $D = M^{-1}AM$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad M = \begin{bmatrix} 1 & 0 & 4 \\ 0 & -\frac{2}{3} & -2 \\ 0 & 1 & 4 \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{bmatrix}$$

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

$$\lambda = 1, 2, 2$$

$$[A - \lambda I] = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 1 & 2 \\ -1 & -5 & -4 \end{bmatrix} \quad \lambda = 2$$

$$x_1 = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$$

$$\begin{aligned} \text{AM} &= 2 \\ \text{GM} &= 1 \end{aligned}$$

$$\text{AM} \neq \text{GM} \quad \text{for } \lambda = 2$$

$\Rightarrow A$ is non diagonalizable.

②

$$A = \begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$$

$$A^0 \rightarrow \lambda$$

If λ is eigen value of matrix A then

① $k\lambda$ is eigen value of kA

② λ^n is eigen value of A^n

③ λ^{-1} is eigen value of A^{-1}

④ $\frac{|A|}{\lambda}$ is eigen value of $\text{adj } A$.

⑤ $f(\lambda)$ is eigen value of $f(A)$

$$\begin{array}{l} \textcircled{10A} \quad \begin{array}{c} 1 \\ \swarrow \\ A \end{array} \begin{array}{c} 2 \\ \searrow \\ A \end{array} \begin{array}{c} 3 \\ \searrow \\ A \end{array} \\ A^3 \quad 20A \\ \begin{array}{c} 1 \\ \swarrow \\ A \end{array} \begin{array}{c} 2 \\ \searrow \\ A \end{array} \begin{array}{c} 3 \\ \searrow \\ A \end{array} \\ A^4 \quad 2^4 \quad 3^4 \end{array}$$

In all above cases eigen vector remain same -

$$IX = \lambda A^{-1}X$$

$$\lambda^{-1}X = A^{-1}X$$

$$X \neq 0$$

$$(kA)X = (k\lambda)X$$

$$AX = \lambda X$$

$$A^2X = A\lambda X$$

$$= \lambda AX$$

$$A^2X = \lambda^2 X$$

eigen values & eigen vectors
 find $A^5, 5A, A^{-1}, \text{adj} A,$
 $A^3 - 3A^2 + 5I, B = I - 6A^{-1}$

$$A = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

as A is Δ form matrix
 $\Rightarrow$ eigen values are $1, 3, -2$

To find $X \neq 0$ s.t. $(A - \lambda I)X = 0$

$$\begin{bmatrix} 1-\lambda & 0 & -3 \\ 0 & 3-\lambda & 2 \\ 0 & 0 & -2-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda = 1 \quad \begin{bmatrix} 0 & 0 & -3 \\ 0 & 2 & 2 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{cases} 2y + 2z = 0 \quad \text{--- (1)} \\ -3z = 0 \Rightarrow z = 0 \\ \text{from (1)} \quad y = 0 \\ \frac{x}{-3} = -\frac{y}{0} = \frac{z}{0} = t \\ x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \end{cases}$$

$$\lambda = 3 \quad \begin{bmatrix} -2 & 0 & -3 \\ 0 & 0 & 2 \\ 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{cases} -2x - 3z = 0 \\ 2z = 0 \Rightarrow z = 0 \text{ & } x = 0 \\ \begin{bmatrix} 0 \\ y \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \end{cases}$$

$$\lambda = -2 \quad \begin{bmatrix} 3 & 0 & -3 \\ 0 & 5 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{cases} 3x + 0 - 3z = 0 \\ 5y + 2z = 0 \\ \frac{x}{-1} = -\frac{y}{5} = \frac{z}{5} = t \\ x_3 = \begin{bmatrix} 5 \\ -2 \\ 5 \end{bmatrix} \end{cases}$$

To find eigen values & vectors of A^5

we know, eigen value of A^5 is λ^5 (if λ is e. value of A)

$\therefore$ e. values of A^5 are $1^5, 3^5, (-2)^5 = 1, 243, -32$
 & corresponding eigen vectors are $x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, x_3 = \begin{bmatrix} 5 \\ -2 \\ 5 \end{bmatrix}$

② $5A$ (if eigen value of A is λ then e. value of kA is $k\lambda$ and e. vector remain same)

$\therefore$ e. values of $5A$ are $5\lambda = 5, 15, -10$ &

& corresponding e. vectors are x_1, x_2, x_3

③ A^{-1} (λ^{-1} is e. value of A^{-1})

$\therefore$ e. values of A^{-1} are $= 1, \frac{1}{3}, \frac{1}{-2}$

& corresponding e. vectors are x_1, x_2, x_3

④ $\text{adj} A$ ($\frac{|A|}{\lambda}$ is e. value of $\text{adj} A$)

$$|A| = -6$$

$\therefore$ eigen values of $\text{adj} A = \frac{-6}{1}, \frac{-6}{3}, \frac{-6}{-2} = -6, -2, 3$

& corr. eigen vectors are x_1, x_2, x_3

⑤ $A^3 - 3A^2 + 5I = f(A)$ [$f(\lambda)$ is e. value of $f(A)$]

$\therefore f(\lambda) = \lambda^3 - 3\lambda^2 + 5$ $\therefore$ eigen values of $f(A)$ are $f(1), f(3), f(-2)$

$$f(1) = 3$$

$$f(3) = 5$$

$$f(-2) = -15$$

$$= 3, 5, -15$$

& corr. e. vectors are x_1, x_2, x_3

$$B = I - 6A^{-1} = f(A)$$

eigen values of $f(A)$ are $f(\lambda)$

$$\therefore f(\lambda) = 1 - 6\lambda^{-1}$$

$$f(1) = 1 - 6 = -5$$

$$f(3) = 1 - \frac{6}{3} = -1$$

$$f(-2) = 1 + \frac{6}{2} = 4$$

$\therefore$ eigen values of $B = I - 6A^{-1}$ are $-5, -1, 4$

& corr. e. vectors are x_1, x_2, x_3

$$A = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

Find eigen values of
 & eigenvectors of
 A^{10} , A^{-1} , $5A$, $\text{adj } A$,
 $A^3 - 3A^2 + 5I$,
 $B = I - 6A^{-1}$

Algebraic Multiplicity (AM)

Geometric Multiplicity (GM) k

If 's'

if there exist non singular matrix M
~~s.t. $B = M^{-1}AM$~~

Diagonalizable

$D = M^{-1}AM$
 M transforming
Modal.

λ is e-value of A

and $AM = GM$ for all
eigen values of A then

A is diagonalisable.

Algebraic Multiplicity (AM)

$$AM = GM$$

for all e-values of A .

of 's'

A, B

B is

similar to A

if

there exist
s.t.

M non singular matrix

$$\textcircled{B} = M^{-1} A M$$

$$D = M^{-1} A M$$

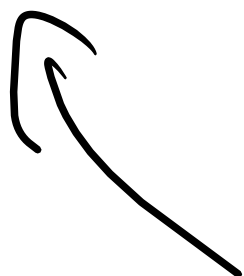
$$M = \begin{bmatrix} x_1 & x_2 \\ \vdots & \vdots \end{bmatrix}$$

$$D = \begin{bmatrix} \textcircled{x_1} & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

x_2
 x_3

$(h - p)M,$

$$g(A) = 0$$



$$\begin{array}{r} f(A) \\ g(\lambda) \overline{) f(\lambda)} \\ \hline \gamma(\lambda) \end{array}$$

$$\lambda^2 A^2 + \lambda_1 A + \lambda_0 I$$

$$\textcircled{\lambda_1} A + \textcircled{\lambda_0} I$$

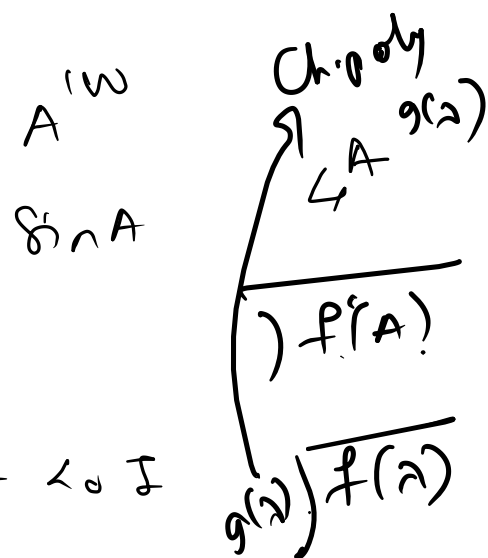
$$f(\lambda) = g(\lambda)q + \gamma(\lambda)$$

$$f(A) = \cancel{g(A)}q + \gamma(A)$$

Functions of square matrix
i.e. $f(A)$

① Case ① For 2×2 matrix
Consider $f(A) = \alpha_1 A + \alpha_0 I$

Case ② For 3×3 matrix
Consider $f(A) = \alpha_2 A^2 + \alpha_1 A + \alpha_0 I$



② Find eigen values of A .

③ Suppose λ is e-value of A .
then find $f(\lambda)$.

Case ① 2×2 matrix

$$f(\lambda) = \alpha_1 \lambda + \alpha_0$$

If two eigen values are distinct then
by substituting these in $f(\lambda)$ we get two
eqns. Solve them & find α_1, α_0 .

Case ② 3×3 matrix

$$f(\lambda) = \alpha_2 \lambda^2 + \alpha_1 \lambda + \alpha_0$$

If all e-values are distinct we get 3 eqns
Solve them & find $\alpha_2, \alpha_1, \alpha_0$.

Note: If eigen values are repeated then find
 $f'(\lambda)$ along with $f(\lambda)$ in order find.
another eqn.

① If $A = \begin{bmatrix} \pi/2 & \pi \\ 0 & 3\pi/2 \end{bmatrix}$ find $\sin A$.

Consider $f(A) = \sin A = \alpha_1 A + \alpha_0 I$ — (1)

Eigen values are $\pi/2, 3\pi/2$

$\therefore f(\lambda) = \sin \lambda = \alpha_1 \lambda + \alpha_0$ — (2)

1.

$\lambda = \pi/2 \Rightarrow 1 = \alpha_1 \pi/2 + \alpha_0$ — (3)

$\lambda = 3\pi/2 \Rightarrow -1 = \alpha_1 3\pi/2 + \alpha_0$ — (4)

$$2 = -\alpha_1 \pi \Rightarrow \boxed{\alpha_1 = -\frac{2}{\pi}}$$

from (3) $1 = -\frac{2}{\pi} \pi/2 + \alpha_0 \Rightarrow \boxed{\alpha_0 = 2}$

$f(A) = \sin A = \alpha_1 A + \alpha_0 I$

$= -\frac{2}{\pi} \begin{bmatrix} \pi/2 & \pi \\ 0 & 3\pi/2 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$= \begin{bmatrix} -1 & -2 \\ 0 & -3 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix}$

$$A = \begin{bmatrix} 2 & 3 \\ -3 & -4 \end{bmatrix}$$

Find A^{100}

-8+9

~~$$A^{99} = 20A + I$$~~

$$f(A) = A^{100} = \lambda_1 A + \lambda_0 I \rightarrow \textcircled{1}$$

$$\text{Ch. eqn } \lambda^2 - (-2)\lambda + 1 = 0$$

$$\lambda^2 + 2\lambda + 1 = 0$$

$$\lambda = -1, -1$$

$$f(\lambda) = \lambda^{100} = \lambda_1 \lambda + \lambda_0 \rightarrow \textcircled{2}$$

$$\lambda = -1 \Rightarrow (-1)^{100} = -\lambda_1 + \lambda_0$$

$$1 = -\lambda_1 + \lambda_0 \rightarrow \textcircled{3}$$

Diff $\textcircled{2}$

wrt λ

for $\lambda = -1$

$$100 \lambda^{99} = \lambda_1$$

$$\Rightarrow 100(-1)^{99} = \lambda_1$$

$$= \boxed{-100 = \lambda_1}$$

from $\textcircled{3}$

$$1 = 100 + \lambda_0 \Rightarrow \boxed{\lambda_0 = -99}$$

$$f(A) = A^{100} = \lambda_1 A + \lambda_0 I = -100 \begin{bmatrix} 2 & 3 \\ -3 & -4 \end{bmatrix} + (-99) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -200 & -300 \\ 300 & 400 \end{bmatrix} + \begin{bmatrix} -99 & 0 \\ 0 & -99 \end{bmatrix} = \begin{bmatrix} -299 & -300 \\ 300 & 301 \end{bmatrix}$$

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \quad \text{find } \underline{e^A} \quad / \quad \textcircled{5^A} \textcircled{HW}$$

$$f(A) = e^A = \lambda_1 A + \lambda_0 I \quad \text{--- (1)}$$

Ch. eqn $\lambda^2 - 6\lambda + 8 = 0 \Rightarrow \lambda = 4, 2$

$$f(\lambda) = e^\lambda = \lambda_1 \lambda + \lambda_0$$

$$\lambda = 4 \quad e^4 = \lambda_1 \cdot 4 + \lambda_0 \quad \text{--- (2)}$$

$$\lambda = 2 \quad e^2 = \lambda_1 \cdot 2 + \lambda_0 \quad \text{--- (3)}$$

$$e^4 - e^2 = 2\lambda_1 \Rightarrow \lambda_1 = \left(\frac{e^4 - e^2}{2} \right)$$

$$\text{from (3)} \quad e^2 = \left(\frac{e^4 - e^2}{2} \right) \cdot 2 + \lambda_0 \Rightarrow \lambda_0 = 2e^2 - e^4$$

$$f(A) = e^A = \lambda_1 A + \lambda_0 I = \left(\frac{e^4 - e^2}{2} \right) \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} + (2e^2 - e^4) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3e^4 - 3e^2}{2} & \frac{e^4 - e^2}{2} \\ \frac{e^4 - e^2}{2} & \frac{3e^4 - 3e^2}{2} \end{bmatrix} + \begin{bmatrix} 2e^2 - e^4 & 0 \\ 0 & 2e^2 - e^4 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{e^4 + e^2}{2} & \frac{e^4 - e^2}{2} \\ \frac{e^4 - e^2}{2} & \frac{e^4 + e^2}{2} \end{bmatrix}$$

since $\frac{3e^4 - 3e^2}{2} + 2e^2 - e^4 = \frac{3e^4 - 3e^2 + 4e^2 - 2e^4}{2} = \frac{e^4 + e^2}{2}$

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Show that

$$e^{At} = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$$

$$f(A) = e^{At} = \alpha_1 A + \alpha_0 I \quad \text{--- (1)}$$

$$\text{Char}^n \quad \lambda^2 - 0\lambda + 1 = 0 \Rightarrow \lambda^2 + 1 = 0 \\ \Rightarrow \lambda = \pm i$$

$$\lambda = i \quad e^{\lambda t} = \alpha_1 \lambda + \alpha_0 \quad \text{--- (2)}$$

$$\lambda = -i \quad e^{-it} = -\alpha_1 i + \alpha_0 \quad \text{--- (3)}$$

$$\begin{array}{r} e^{it} + e^{-it} = 2\alpha_0 \Rightarrow \alpha_0 = \frac{e^{it} + e^{-it}}{2} \\ = \cos t \end{array}$$

$$\begin{array}{r} \text{(2)} - \text{(3)} \\ e^{it} = \alpha_1 i + \alpha_0 \\ - e^{-it} = -\alpha_1 i + \alpha_0 \\ \hline e^{it} - e^{-it} = 2\alpha_1 i \\ \Rightarrow \alpha_1 = \frac{e^{it} - e^{-it}}{2i} = \sin t \end{array}$$

$$e^{At} = \alpha_1 A + \alpha_0 I = \sin t \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} + \cos t \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$$

$$\cancel{e^{At}} = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 4 \\ 2 & 1 \end{bmatrix} \quad p-T. \quad 3 \tan A = A \tan 3.$$

$$f(A) = 3 \tan A = \lambda_1 A + \lambda_0 I \quad \text{--- (1)}$$

$$\lambda^2 - 0\lambda + (-9) = 0 \quad \Rightarrow \quad \lambda^2 - 9 = 0$$

$$\lambda = \pm 3$$

$$f(\lambda) \Rightarrow 3 \tan \lambda = \lambda_1 \lambda + \lambda_0$$

$$\lambda = 3 \quad 3 \tan 3 = \lambda_1 \cancel{3} + \lambda_0 \quad \text{--- (2)}$$

$$\lambda = -3$$

$$-3 \tan 3 = -\lambda_1 \cancel{3} + \lambda_0$$

$$\tan(-3)$$

$$= -\tan 3$$

$$0 = 2\lambda_0 \Rightarrow \lambda_0 = 0$$

$$\text{from (2)} \quad \cancel{3} \tan 3 = \lambda_1 \cancel{3}$$

$$\boxed{\lambda_1 = \tan 3}$$

$$3 \tan A = \lambda_1 A + \lambda_0 I$$

$$= (\tan 3) A + 0$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

find
 A^{50}

find $A^7 - 9A^2 + I = f(A)$

$$A = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}$$

(Ch. 29^n)

$$g(\lambda) = 0$$

$$g(A) = 0$$

$$f(A) = \frac{g(A)h(A)}{+v(A)}$$

$$= \lambda_1 A + \lambda_0 I$$

$$\lambda_1 = 329$$

$$\lambda_0 = 520$$

Consider,

$$f(A) = A^{50} = \lambda_2 A^2 + \lambda_1 A + \lambda_0 I \quad \text{--- (1)}$$

$$\text{Ch. } 29^n \quad \lambda^3 - \lambda^2 + [0 - 1]\lambda + 1 = 0$$

$$\lambda^3 - \lambda^2 - \lambda + 1 = 0$$

$$\lambda = -1, 1, 1$$

$$f(\lambda) = \lambda^{50} = \lambda_2 \lambda^2 + \lambda_1 \lambda + \lambda_0 \quad \text{--- (2)}$$

from (2)

$$\lambda = -1 \Rightarrow 1 = \lambda_2 - \lambda_1 + \lambda_0$$

$$\lambda = 1 \Rightarrow 1 = \lambda_2 + \lambda_1 + \lambda_0$$

$$\text{Diff (2) wrt } \lambda \Rightarrow 50\lambda^{49} = 2\lambda_2 \lambda + \lambda_1$$

$$\Rightarrow 50 = 2\lambda_2 + \lambda_1$$

$$\lambda_2 - \lambda_1 + \lambda_0 = 1$$

$$\lambda_2 + \lambda_1 + \lambda_0 = 1$$

$$2\lambda_2 + \lambda_1 = 50$$

$$\lambda_2 = 25$$

$$\lambda_1 = 0$$

$$\lambda_0 = -24$$

$$f(A) = A^{50} = \lambda_2 A^2 + \lambda_1 A + \lambda_0 I = 25 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + 0 + (-24) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

HW

Derogatory & Non Derogatory Matrix.

* Monic Poly^n :- highest deg. coeff. is 1

eg. Ch. poly^n , $\lambda^3 - 2\lambda^2 + 1 = 0$

* Minimal poly^n :- annihilates matrix A .

Note $f(\lambda)$ is ch. poly^n of matrix A
then we know that $f(A) = 0$
— (H. Thm.)

Note

Min $\text{poly}^n \neq \text{Ch. poly}^n \Rightarrow$ derogatory

Min $\text{poly}^n = \text{Ch. poly}^n \Rightarrow$ Non-derogatory

8. If A is derogatory & find its minimal polyⁿ

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

$$\text{Ch. eqⁿ } |A - \lambda I| = 0$$

$$\lambda^3 - 12\lambda^2 + [(18 + 9 + 18) - (4 + 4 + 1)]\lambda - |A| = 0$$

$$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$$\lambda = 8, 2, 2$$

$$\therefore (\lambda - 8)(\lambda - 2)(\lambda - 2) = 0$$

$$\text{Consider, } (\lambda - 8)(\lambda - 2) = \lambda^2 - 10\lambda + 16$$

$$\text{Consider } A^2 - 10A + 16I$$

$$= \begin{bmatrix} 44 & -20 & 20 \\ -20 & 14 & -10 \\ 20 & -10 & 14 \end{bmatrix} - \begin{bmatrix} 60 & -20 & 20 \\ -20 & 30 & -10 \\ 20 & -10 & 30 \end{bmatrix} + \begin{bmatrix} 16 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 16 \end{bmatrix}$$
$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

As $\lambda^2 - 10\lambda + 16$ annihilates A
 $\therefore$ minimal polyⁿ = $\lambda^2 - 10\lambda + 16 \Rightarrow \underline{\underline{A \text{ is derogatory}}}$

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & +3 \end{bmatrix}$$

$$\lambda^3 - 18\lambda^2 + 45\lambda - 0 = 0$$

$$\lambda(\lambda^2 - 18\lambda + 45) = 0$$

$$\lambda = 0, 3, 15$$

as all e. values are distinct

$\Rightarrow$ min poly = ch. polyⁿ

$\Rightarrow$ A is nonderogatory

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

e.v. 1, 1, 1

$$\text{Ch. eq}^n (\lambda - 1)(\lambda - 1)(\lambda - 1) = 0$$

Concl^{us} $\lambda = 1$

$$A - I = I - I = 0$$

$$\text{min poly}^n \lambda - 1$$

SVD

$$A = U_{m \times m} D_{m \times n} V^T_{n \times n} \quad m \times n \quad \begin{bmatrix} x_1 & x_2 & x_3 \\ | & | & | \\ | & | & | \\ | & | & | \end{bmatrix}$$

where $U \rightarrow$ orthogonal matrix

$D \rightarrow$ Diagonal matrix

$V \rightarrow$ Orthogonal matrix.

① Find $A^T A$

② Find eigen values & eigen vectors of $A^T A$

③ An

say $\lambda_1, \lambda_2, \lambda_3$, $\lambda_1 > \lambda_2 > \lambda_3$

$$D = \begin{bmatrix} \sqrt{\lambda_1} & 0 & 0 \\ 0 & \sqrt{\lambda_2} & 0 \\ 0 & 0 & \sqrt{\lambda_3} \end{bmatrix}$$

④ Suppose x_1, x_2, x_3 are corr. e. vectors.

then make all these e. vectors unit vectors.

$$\therefore V = \begin{bmatrix} \frac{x_1}{|x_1|} & \frac{x_2}{|x_2|} & \frac{x_3}{|x_3|} \end{bmatrix}$$

clearly V is orthogonal.

⑤

suppose $U = [u_1, u_2, u_3]$
 u_1, u_2, u_3 are columns of U

v_i 's are columns of V

$$u_i = \frac{1}{\sqrt{\lambda_i}} A v_i$$

$$A = U D V^T \quad A v = U D V^T v$$

$$A v_i = U D \quad A v_i = u_i \sqrt{\lambda_i}$$

we require u_i 's as orthogonal unit vectors.

or
Find eigen vectors of $A^T A$

suppose u_1, u_2, u_3 are e.v. of $A^T A$

$$\therefore U = \begin{bmatrix} \frac{u_1}{|u_1|} & \frac{u_2}{|u_2|} & \frac{u_3}{|u_3|} \end{bmatrix}$$

$$A = U D V^T$$

$$A v = U D V^T v$$

$$A v_i = u_i \sqrt{\lambda_i}$$

Find $A^T A$

Also find e. values & e. vectors of $A^T A$

Suppose v_1, v_2, v_3 are e. vectors of $A^T A$

$$\text{then } V = \begin{bmatrix} \frac{v_1}{|v_1|} & \frac{v_2}{|v_2|} & \frac{v_3}{|v_3|} \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} \rightarrow A_{2 \times 2} = U_{2 \times 2} D_{2 \times 2} V^T_{2 \times 2}$$

$$B = A A^T = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 32 & 0 \\ 0 & 18 \end{bmatrix}$$

eigen values are 32, 18

$$D = \begin{bmatrix} \sqrt{32} & 0 \\ 0 & \sqrt{18} \end{bmatrix} \quad \left| \begin{array}{l} u = (x, y) \\ |u| = \sqrt{x^2 + y^2} \end{array} \right.$$

to find $x \neq 0$ s.t. $(B - \lambda I)x = 0$

$$\begin{bmatrix} 32 - \lambda & 0 \\ 0 & 18 - \lambda \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\lambda = 32 \quad \begin{bmatrix} 0 & 0 \\ 0 & -14 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow y = 0 \quad \text{s.t.} \quad \begin{pmatrix} x \\ 0 \end{pmatrix}$$

$$\lambda = 18 \quad \begin{bmatrix} 14 & 0 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow x = 0 \quad \text{s.t.} \quad \begin{pmatrix} 0 \\ y \end{pmatrix}$$

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$C = A^T A = \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 25 & 7 \\ 7 & 25 \end{bmatrix} \checkmark$$

$$\lambda^2 - 50\lambda + 576 = 0 \Rightarrow \lambda = 32, 18$$

to find $x \neq 0$ s.t. $(C - \lambda I)x = 0$

$$\begin{bmatrix} 25 - \lambda & 7 \\ 7 & 25 - \lambda \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \left| \begin{array}{l} \lambda = 32 \quad \begin{bmatrix} -7 & 7 \\ 7 & -7 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad 7x - 7y = 0 \\ \text{s.t.} \quad \begin{pmatrix} x \\ y \end{pmatrix} \quad x_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad |x_1| = \sqrt{2} \end{array} \right.$$

$$\lambda = 18 \quad \begin{bmatrix} 7 & 7 \\ 7 & 7 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow x = -y \quad \frac{x_1}{|x_1|} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$x_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \Rightarrow x_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad |x_2| = \sqrt{2}$$

$$\frac{x_2}{|x_2|} = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \quad V = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$A = U D V^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{32} & 0 \\ 0 & \sqrt{18} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} \quad (\text{HW})$$

$$A = \begin{pmatrix} 2 & 0 \\ -3 & 2 \end{pmatrix}$$

$$B = A A' = \begin{pmatrix} 4 & -6 \\ -6 & 13 \end{pmatrix}$$

$$\lambda^2 - 17\lambda + 16 = 0$$

$$\lambda = 16, 1$$

$$\lambda = 16, 1$$

$$6x + 3y = 0$$

$$y = -2x$$

$$x_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$|x_1| = \sqrt{5}$$

$$\begin{pmatrix} x \\ -2x \end{pmatrix}$$

$$x_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$|x_2| = \sqrt{5}$$

$$V = \begin{pmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ -2/\sqrt{5} & 1/\sqrt{5} \end{pmatrix}$$

$$D = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$$

$$C = A' A = \begin{pmatrix} 13 & -6 \\ -6 & 4 \end{pmatrix}$$

$$\lambda = 16, 1$$

$$(C - \lambda I) x = 0$$

$$\begin{pmatrix} 13-\lambda & -6 \\ -6 & 4-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\lambda = 16$$

$$-3x - 6y = 0 \Rightarrow$$

$$x = -2y$$

$$x_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$|x_1| = \sqrt{5}$$

$$\lambda = 1$$

$$12x - 6y = 0 \Rightarrow$$

$$y = 2x$$

$$x_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$|x_2| = \sqrt{5}$$

$$V = \begin{pmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{pmatrix}$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$1+y-x$$

$$A = U_{3 \times 2} D_{3 \times 2} V_{2 \times 2}^T$$

$$B = AA^T = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\det(B - \lambda I) = 0 \Rightarrow \lambda^3 - 6\lambda^2 + [(4+4+4)-4]\lambda - 0 = 0$$

$$\lambda^3 - 6\lambda^2 + 8\lambda = 0 \Rightarrow \lambda(\lambda^2 - 6\lambda + 8) = 0$$

$$\Rightarrow \lambda = 0, 2, 4$$

$$\lambda = 4, 2, 0$$

$$\exists x \neq 0 \text{ s.t. } (B - \lambda I)x = 0 \quad \begin{bmatrix} 2-\lambda & 2 & 0 \\ 2 & 2-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda = 4 \quad \begin{bmatrix} -2 & 2 & 0 \\ 2 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2x - 2y = 0 \Rightarrow x = y \quad \text{sum} \begin{bmatrix} y \\ y \\ 0 \end{bmatrix}$$

$$z = 0$$

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad |x_1| = \sqrt{2}, \quad \frac{x_1}{|x_1|} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$$

$$\lambda = 2, \begin{bmatrix} 0 & 2 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$y = 0 \quad x = 0, \quad \text{sum} \begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad |x_2| = 1$$

$$\lambda = 0 \quad \begin{bmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow 2x + 2y = 0 \Rightarrow x = -y \quad \text{sum} \begin{bmatrix} -y \\ y \\ 0 \end{bmatrix}$$

$$z = 0$$

$$x_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \quad |x_3| = \sqrt{2}, \quad \frac{x_3}{|x_3|} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$$

$$U = \left[\frac{x_1}{|x_1|} \quad \frac{x_2}{|x_2|} \quad \frac{x_3}{|x_3|} \right]$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}_{3 \times 2}$$

$$C = A^T A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\text{To find } x \neq 0 \text{ s.t. } [C - \lambda I]x = 0 \quad \lambda^2 - 6\lambda + 8 = 0 \Rightarrow \lambda = 4, 2$$

$$\lambda = 4 \quad \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x = y \quad \text{sum} \begin{bmatrix} y \\ y \end{bmatrix} \quad x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad |x_1| = \sqrt{2}$$

$$\frac{x_1}{|x_1|} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\lambda = 2 \quad \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x = -y \quad \text{sum} \begin{bmatrix} -y \\ y \end{bmatrix} \quad x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad |x_2| = \sqrt{2}$$

$$V = \left[\frac{x_1}{|x_1|} \quad \frac{x_2}{|x_2|} \right] = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad \frac{x_2}{|x_2|} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = U D V^T$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ -3 & 2 \end{bmatrix}$$